

DX Assistant

V0.14.1 Beta User Manual

Receive-only FT8 target monitoring and safe band searching

Safety boundary DX Assistant never initiates transmission. It never controls PTT, Enable Tx, Halt Tx, radio mode, split state, or RF power. The operator always remains in control.

For WSJT-X 3.0.1 and capability-compatible OmniRig Rig 1 profiles

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Beta software - verify all radio and amplifier settings before transmitting

1. Quick start

Before starting Place the radio in split operation with receive on VFO A and transmit on VFO B. Confirm a safe RF power setting manually. Start OmniRig and WSJT-X before beginning the band search.

1. Extract the complete portable ZIP. Keep DXAssistant.exe, config.template.json, and the _internal folder together. The first launch creates the private local config.json.
2. In WSJT-X, configure UDP reporting to 127.0.0.1 port 2237 and enable Settings > Reporting > Accept UDP requests.
3. Configure OmniRig Rig 1 with the exact profile for the connected radio and confirm that OmniRig can read it.
4. Start DXAssistant.exe and select Check OmniRig compatibility before attempting a band search.
5. Start DXAssistant.exe, choose Change target while stopped, and enter the DX callsign.
6. Optional: while stopped, choose Mobile alerts, enter your Pushover credentials, enable mobile alerts, and send a test notification.
7. Select Start monitoring and wait until the WSJT-X version, frequency, and mode appear.
8. Choose the required dwell time, then select Start band search.
9. When the target is decoded, read the latest target decode, acknowledge the alert, and decide whether to call in WSJT-X.

The two actions people most often confuse

Control	Meaning
Acknowledge target alert	I have seen the notification. It stops the current alert but leaves the radio on the detected band and retains the decodes.
Resume search	Continue hunting. It safely moves both VFOs to the next enabled band's configured base frequency and restarts the dwell cycle.

Normal target workflow

- Want to call: acknowledge the alert, check the target's period in the latest decode, then use WSJT-X normally.
- Do not want to call: acknowledge the alert, then select Resume search.

- After a QSO: release Enable Tx in WSJT-X, acknowledge any outstanding alert, then select Resume search.

2. Installation and prerequisites

Portable package

DX Assistant V0.14.1 is supplied both as a portable ZIP and as an unsigned per-user Windows installer. Python is bundled; a separate Python installation is not required. The installer adds Start Menu and optional desktop shortcuts and preserves the operator-owned config.json during upgrades and uninstall.

- Send or retain the complete ZIP file, not DXAssistant.exe alone.
- Use Extract All before running the program. Do not run it from inside the ZIP preview.
- Keep the _internal folder beside DXAssistant.exe.
- Keep config.template.json beside DXAssistant.exe. On first launch, DX Assistant creates config.json beside it; this private live file stores persistent operating choices and optional Pushover keys.
- Close an older DX Assistant instance before starting the new one, otherwise UDP port 2237 may already be in use.

Required companion software

Component	Purpose	Required setup
WSJT-X 3.0.1	Supplies live Status, Heartbeat, and Decode UDP messages.	UDP 127.0.0.1:2237; Accept UDP requests enabled.
OmniRig 1.20	Translates the guarded VFO frequency requests for the configured radio.	Rig 1 online with the exact radio profile.
Compatible radio	Provides the required dual-VFO and safety-state capabilities through OmniRig.	Split on; receive A / transmit B; radio in RX.

Experimental radio compatibility DX Assistant has no radio-model allow-list. It uses only the OmniRig API and requires readable/writable VFO A and B frequencies plus readable VFO, split, receive and transmit state. A positive compatibility check identifies a candidate profile, not a validated radio. Follow the colleague test guide with the amplifier in standby.

WSJT-X reporting setup

1. Open WSJT-X Settings and select Reporting.
2. Set the UDP Server to 127.0.0.1 and the UDP Server port number to 2237.
3. Select Accept UDP requests. Accepted UDP request restores window is optional.
4. Start WSJT-X Monitor. DX Assistant should then show the WSJT-X version, dial frequency, and FT8 mode.

3. Understanding the dashboard

Area	What it tells you
Monitoring status	Overall state, selected target, WSJT-X connection, dial frequency in MHz, mode, and total/target decode counts.
Session band plan	Enabled antenna bands and saved FT8 base frequencies. The current band is highlighted green.
Current band	The band identified from WSJT-X Status plus the read-only WSJT-X transmit audio offset.
Band search	Dwell time, search state, countdown, start/pause/resume control, and persistent OmniRig interlock errors.
PSK Reporter	Your locator/radius and any evidence-based band focus. It falls back to a full enabled-band sweep when reports are unavailable.
Recent WSJT-X activity	The latest 12 local decodes in chronological order. The display scrolls upward and keeps the newest decode visible at the bottom.
Target decodes	All target decodes retained for the current session, including time, band, signal report, offset, and message.

Colour cues

Colour	Meaning
Green	An available start, resume, or Change target action, or the current enabled band.
Amber/yellow	An available stop, pause, acknowledge, or stopped-only configuration action.
Grey	The control is not currently available.
Red text	A saved non-standard operating frequency, or target-alert emphasis.

The wording and enabled/disabled state remain authoritative; colour is an additional cue only.

Monitoring states

State	Meaning
Stopped	No WSJT-X listener is active. Configuration controls are available.
Starting	The local UDP listener is opening.
Monitoring	WSJT-X messages are being received; band search may be stopped or active.
Target decoded	The target has been heard locally and the alert awaits acknowledgement.
Degraded	WSJT-X data timed out; the app waits for automatic recovery.
Error	The listener could not start or another non-recoverable startup problem occurred.

4. Starting and controlling a search

Start monitoring first

Start monitoring opens the local WSJT-X UDP listener. It does not tune the radio. Once the first safe receive-only WSJT-X Status packet arrives, DX Assistant automatically sends the selected target to WSJT-X and asks it to generate standard messages.

No automatic transmitting Preparing the DX Call in WSJT-X does not select Enable Tx, choose an even/odd period, reply to a decode, or transmit.

Start band search

Band searching is a separate explicit action. Choose a dwell of 60, 90, 120, 180, 240, or 300 seconds. The default is 120 seconds so complete 15-second FT8 intervals have a practical chance to be decoded.

Before every tune, the OmniRig bridge checks that Rig 1 exposes the complete required capability set, is online, the radio is receiving, split is on, and routing is receive-A/transmit-B. It moves VFO B first, then VFO A, and verifies both at the configured base frequency.

Pause, resume, and immediate-band monitoring

- Pause search stops future hops but continues WSJT-X monitoring.
- Resume search realigns both VFOs and continues the search only when WSJT-X is not transmit-enabled.

- Monitor selected band now immediately aligns both VFOs to a selected enabled band, clears the blue selection/focus highlight, begins a fresh dwell there, and then continues round-robin searching. The green current-band indication remains.
- If WSJT-X Enable Tx becomes active, DX Assistant hands VFO control to the operator and leaves the search paused until Enable Tx is released and Resume search is explicitly selected.

Changing the target

1. Select Stop.
2. Select Change target and enter one callsign without spaces.
3. Select Start monitoring. The new target is automatically prepared in WSJT-X when a safe Status packet is received.

Re-sending the target

Use Re-send target to WSJT-X if WSJT-X did not accept the first request or if you want to send the target again. When a target grid has been decoded, the re-send can include it; otherwise the existing WSJT-X DX Grid is left unchanged.

5. When the target is decoded

A matching decode means the wanted station was heard locally. A callsign merely appearing as the station being called is not treated as proof that the target itself was heard.

What happens automatically

- The state changes to Target decoded.
- The local bell sounds and the visible DX Assistant window is raised once.
- A snapped or maximized window keeps its layout; only a truly minimized window is restored.
- Band hopping enters Target hold and remains on the detected band.
- The decode is added to the retained Target decodes panel.
- Further target decodes accumulate without repeatedly ringing or moving the window until the alert is acknowledged.

Operator decision

Read the latest decode The DX station can change time slot. Confirm the current period from the retained target decode before selecting Enable Tx in WSJT-X. The practical default for many DXpeditions is 00/30, but DX Assistant does not assume or set it.

1. Select Acknowledge target alert to confirm that you have seen the notification.
2. If calling, work entirely in WSJT-X. Select Enable Tx only after checking period, audio offset, radio power, and amplifier conditions.
3. When finished, release Enable Tx in WSJT-X.
4. Select Resume search when you want DX Assistant to continue band hopping.

What acknowledgement does not do

- It does not erase target decodes or reset decode counters.
- It does not enable transmission or change radio settings.
- It does not resume the band search.
- It does not prevent you from continuing to monitor the detected band.

6. Configuring the band plan

Band-plan controls are intentionally available only while monitoring is stopped. Select a row before editing it.

Control	Effect	Persistence
Edit frequency	Sets a three-decimal MHz base frequency within the selected amateur band. Non-standard values appear red.	Saved across restarts.
Enable / disable	Includes or excludes the band to match the connected antenna.	Saved across restarts.
Restore frequencies	Restores standard FT8 base frequencies without changing antenna enablement.	Saved immediately.

Supplied band plan

Band	Standard FT8 MHz
160m	1.840
80m	3.573
40m	7.074
30m	10.136
20m	14.074
17m	18.100
15m	21.074
12m	24.915
10m	28.074
6m	50.313

160m, 80m and 6m start disabled. Enable them only after confirming antenna coverage.

Power reminder DX Assistant has no power display or power-setting control because it cannot set or verify the radio's RF power through the OmniRig API used here. Select and confirm a safe setting directly at the radio. The operator remains responsible for the radio and amplifier.

WSJT-X transmit audio offset

The Current band panel displays the WSJT-X Tx offset in hertz. This value is read-only and is not added to VFO B. WSJT-X combines its audio offset with the dial frequency and manages split-frequency translation when required.

7. PSK Reporter band focus

While monitoring, DX Assistant checks recent FT8 reports for the selected target no more frequently than once every five minutes. Reports are used only to rank bands; PSK Reporter never controls the radio.

Setting the search area

1. Enter your current Maidenhead locator in Area, for example JO03.
2. Choose a radius of 500, 1,000, 1,500, 2,500, or 5,000 km.
3. Select Apply. The locator and radius are saved for future sessions and a fresh query is requested.

How focus works

- Only recent FT8 reports from receivers within the selected radius are considered.
- Repeated reports from the same receiving station on one band count once.
- Exact reported dial/audio differences are ignored; only the amateur band matters.
- Reported enabled bands are ranked using receiver count, proximity, and recency.
- A changed focus begins at the next dwell boundary so an FT8 interval is not cut short.
- If no useful report is available, DX Assistant automatically sweeps every enabled band.

Large report counts A display such as Focus 20m (337) means 337 distinct nearby receiving stations reported the target on 20m during the query window. It does not mean 337 target transmissions.

8. Pushover mobile alerts

Phone requirement The Pushover app is required on the phone. Mobile alerts are optional and do not replace the local audible and visual target alert.

Pushover setup

1. Install the Pushover app on the phone, create a Pushover account, and complete its device setup.
2. Sign in at pushover.net. Copy the 30-character User Key shown on the dashboard.
3. Create an application/API token for DX Assistant at pushover.net/apps and copy its 30-character API Token.
4. Stop DX Assistant monitoring, then select Mobile alerts in the band-plan controls.
5. Enter the User Key and API Token, select Enable mobile alerts for target detections, then choose Send test notification.
6. Confirm the test arrives on the phone. Choose Save to retain the settings.

Keep credentials private The keys are masked in the GUI and stored only in the local config.json. They are never written to DX Assistant logs and the supplied package contains blank fields. Treat a configured config.json as private.

Notification behaviour

- DX Assistant makes up to three normal-priority delivery attempts when a target alert is newly raised. Because a network timeout can occur after acceptance, a rare duplicate phone notification is possible.
- The message includes callsign, band, dial frequency, mode, UTC decode time, and SNR.
- Repeated decodes do not repeatedly notify the phone while the same target alert remains active.
- Acknowledging an alert does not control the phone or radio; a later newly raised alert can send another message.
- Internet, credential, quota, or Pushover failure never prevents the local alert or monitoring.
- Diagnostics shows only generic sent/failed status and never displays the keys.

Notification only Pushover adds no PTT, reply, radio-mode, split, power, or WSJT-X transmit command. DX Assistant remains receive-only and the operator remains in control.

9. Troubleshooting

Symptom	Likely cause	Action
WSJT-X shows Not listening	Monitoring has not started.	Select Start monitoring.
No WSJT-X version/frequency appears	UDP reporting is not reaching port 2237.	Check 127.0.0.1:2237, start WSJT-X Monitor, and confirm only one DX Assistant is running.
Listener error 10048	Another program or DX Assistant instance already owns port 2237.	Close the other instance, then start monitoring again.
Search blocked: split/routing/RX error	The radio state failed a safety interlock.	Confirm split on, receive A / transmit B, radio in RX, and OmniRig Rig 1 online; then select Resume search.
Decodes roll but band search says error	WSJT-X UDP is healthy, but OmniRig tuning was blocked.	Read the lasting error in the Band search panel; do not treat incoming decodes as proof that CAT is ready.
DX Call was not filled	WSJT-X is not accepting UDP requests, or was transmit-enabled.	Enable Accept UDP requests, release Enable Tx, then select Re-send target to WSJT-X.
PSK Reporter unavailable	No nearby recent reports or the service/query is unavailable.	No action is essential; the app continues a full enabled-band sweep.
Pushover test fails	The phone app, internet connection, User Key, API Token, or Pushover service/quota is unavailable.	Check the app/account and both 30-character keys, then try Send test notification again. Local alerts continue normally.
Search will not resume	Enable Tx remains selected, an alert is unacknowledged, or an interlock still fails.	Release Enable Tx, acknowledge the alert, correct any displayed interlock error, then select Resume search.
EXE fails after being moved	DXAssistant.exe was separated from _internal or config.template.json.	Extract and run the complete portable folder.

Logs and retained information

The two on-screen decode tables are session displays. Clear displays removes their visible rows but does not reset the counters. Target decodes are retained on screen for the current session, and received decodes are also written to daily CSV logs. State changes, operator actions, tuning attempts, failures and target events are written to append-only UTC JSON Lines event logs. If a log write fails, up to 250 audit records are buffered in memory and flushed after writing recovers.

Diagnostics shows the application version, target, WSJT-X state, current band, search, PSK Reporter and mobile-alert status, configuration and log locations, and audit-buffer health. It never shows Pushover

credentials. It also provides Test alarm and Open log folder actions. Mute 15 min suppresses audible bells temporarily while visual and optional phone alerts remain active.

10. Sharing, upgrading, and safety reference

Sending the Beta to another operator

1. Send either the complete DXAssistant-v0.14.1-beta-Windows-portable.zip file or the unsigned DXAssistant-v0.14.1-beta-Setup.exe installer.
2. Ask the recipient to extract the ZIP and run DXAssistant.exe from the folder containing _internal and config.template.json. DX Assistant creates the recipient's private config.json on first launch.
3. The recipient must configure their own WSJT-X UDP settings, OmniRig Rig 1, antenna band selection, locator/radius, operating frequencies, optional Pushover account/keys, and a safe RF power setting directly at the radio.
4. Include COLLEAGUE_TEST_GUIDE.md. Ask the recipient to run the read-only compatibility check first and perform the controlled two-band test with the amplifier in standby.

Upgrading

- Close the running old version before opening the new version.
- Extract each release into its own folder so rollback remains easy.
- To carry compatible settings forward, make a backup and then copy the previous config.json beside the new DXAssistant.exe before starting it.
- Never copy only the new EXE over an old _internal folder; keep each packaged build together.

Non-negotiable operator checks

Before Enable Tx Confirm the callsign, latest decoded period, audio offset, radio split/routing, selected RF power, amplifier limits, antenna suitability, and that the operating frequency is legal for your licence and location.

- DX Assistant does not initiate or halt transmission.
- DX Assistant does not choose even/odd periods.
- DX Assistant does not set or verify RF power.
- DX Assistant does not change radio mode or split state.

- Its OmniRig bridge writes only verified VFO A and VFO B frequencies during an operator-started search.
- Releasing Enable Tx is a handover signal, not an automatic instruction to resume searching; Resume search remains explicit.

Control summary

Control	Available when	Result
Start monitoring	Stopped	Starts WSJT-X UDP reception; does not tune the radio.
Stop	Monitoring	Stops monitoring and band search.
Change target	Stopped	Sets the callsign to hunt.
Acknowledge target alert	Alert active	Marks the alert as seen; search remains held.
Re-send target to WSJT-X	Live receive-only WSJT-X Status	Fills DX Call and asks WSJT-X to generate messages.
Clear displays	Any time	Clears visible decode rows; counters remain.
Mobile alerts	Stopped	Configures optional Pushover phone notification and sends a test message.
Start/Pause/Resume search	Monitoring and receive-safe	Controls only the explicit guarded band-search cycle.
Monitor selected band now	Monitoring and receive-safe	Tunes both VFOs to the selected enabled band, clears blue selection, and starts a fresh dwell.

End of manual